

C O N C E P T N O T E

AI-Enabled Research Infrastructure at KIPRE

Embracing Agentic AI to Digitalize Preclinical and Human Sample Research

Using Malaria as the Pilot Study

Integrated with a Technical Implementation Roadmap

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June 2026

C O N F I D E N T I A L — F O R I N T E R N A L R E V I E W

*How to read this document: unhighlighted text is the original concept note. **Text like this** marks edits to the original wording; **text like this** marks content integrated from a proposed Technical Implementation Roadmap.*

1. Background and Context

The Kenya Institute of Primate Research (KIPRE) is East Africa’s premier biomedical and preclinical research institution, operating under the Ministry of Health. Its mandate to promote public health through innovative preclinical and biomedical research, training, and biodiversity conservation places it at the intersection of science, public health, and policy.

Despite its scientific strengths, KIPRE’s operational infrastructure currently relies heavily on manual, paper-based, and fragmented digital systems. Sample tracking, reagent inventory, animal request approvals, data records, and research outputs are managed across disparate tools and formats, creating significant risks: data inconsistency, audit failures, reagent wastage, and bottlenecks that slow the pace of discovery.

At the same time, Kenya’s National AI Strategy 2025–2030 calls explicitly for AI adoption within public health and research institutions, and global funders including NIH, the Gates Foundation, and Wellcome Trust increasingly require FAIR (Findable, Accessible, Interoperable, Reusable) data practices as a condition of funding.

Opportunity

The convergence of AI maturity, national AI policy, and KIPRE’s research complexity creates a unique window to implement digitalization and deploy agentic AI autonomous systems.

In May 2026, KIPRE’s Director General, Dr. Peter Mwethera and AI Champion, Dr. Faith I. Onditi, hosted representatives of the Upanzi Network from Carnegie Mellon University Africa for a strategic meeting and laboratory tour. The Upanzi Network is CMU-Africa’s Digital Public Infrastructure research lab, with active deployments of agentic AI in cybersecurity, healthcare reconciliation, and national identity systems across the continent.

Following this meeting and lab tour, both institutions agreed to move forward with a formal collaboration to build and deploy an AI-enabled research infrastructure at KIPRE, using malaria research as the initial pilot.

2. Problem Statement

KIPRE’s current operational model faces five interconnected challenges that constrain research quality, efficiency, and compliance:

Challenge Area	Current Impact
Fragmented and Siloed data systems	Samples, inventory and results data held in separate, incompatible tools and paper records
Manual reconciliation processes	High error rates, compliance risks, and inability to generate real-time audit trails for funders

Challenge Area	Current Impact
No sample lifecycle tracking	Lost, mislabelled, or untraceable samples; broken chain of custody for biospecimens
Data integrity gaps	Inconsistencies between lab activity, inventory consumption, and financial records
Limited analytical capacity	Time-consuming manual data cleaning delays publication-ready outputs by weeks per study

These challenges directly undermine KIPRE’s ability to compete for international research funding, meet ISO and GLP standards, and fulfil its mandate as a regional centre of excellence.

3. Proposed Solution : AI-Enabled Research Infrastructure

This concept proposes the development and deployment of a modular, open-source, agentic AI system that digitalizes KIPRE’s five core operational domains. Rather than implementing a conventional static software system, the proposed approach uses ‘agentic AI’, an autonomous agents that actively monitor, correlate signals, and flag anomalies within strict human-oversight guardrails consistent with KIPRE’s ethics framework and ISERC governance.

The system is delivered in two clearly sequenced stages. Stage one is digitization — replacing the manual, paper-based and fragmented-digital processes of Section 2 with structured digital capture, barcode-tracked samples, digital workflows and a single FAIR-compliant repository. Stage two is agentic augmentation — layering the autonomous agents on top of that digital foundation. This ordering is deliberate: agents can only monitor, correlate and flag once the underlying processes are digital and producing clean, structured, timestamped data, so each pillar is digitized and validated first and the agent layer is switched on only afterwards. Digitization alone already meets several pilot KPIs (sample traceability, data-entry error reduction, audit readiness); the agentic layer then adds the harder gains — reagent-wastage forecasting, anomaly detection and faster sample-to-insight — and the platform is fully useful even if the agentic layer is deferred.

3.1 The Five Digital Pillars

Pillar 1: Inventory Management (Samples & Reagents)

A real-time digital inventory system tracking all biological samples, reagents, consumables, and equipment. Key features include barcode/QR-code assignment at point of receipt, automated low-stock and expiry alerts, chain-of-custody logging from acquisition to disposal, and integration with procurement workflows. AI agents continuously monitor stock levels and flag discrepancies between recorded and physical inventory.

Pillar 2: Animal Request & Ethics Workflow

End-to-end digitization of animal use requests, ISERC ethics approvals, scheduling, and allocation records. The system replaces paper-based request forms with a mobile-accessible digital workflow, featuring automated routing to approvers, status notifications, and a complete, auditable approval trail. This directly addresses compliance requirements for preclinical research funders.

Pillar 3: Work Orders & Sample Collection

A barcode-linked work order system that tracks each sample from collection through processing, assaying, and archiving. Technicians log check-ins at every handoff point via mobile device, creating an unbroken chain of custody. AI agents flag samples that miss expected processing windows or show anomalous readings at any stage.

Pillar 4 : Data Records & Storage

A centralized, FAIR-compliant research data repository replacing fragmented spreadsheets and paper ledgers. The system features role-based access control, version history, automated backups, and full audit logging. Data entry is structured and validated at point of capture, eliminating the downstream cleaning burden that currently delays analysis.

Pillar 5: Data Analysis & Results Output

An integrated analytics pipeline connecting raw assay data to visualized, publication-ready outputs. An agentic AI layer flags statistical anomalies, suggests quality-control actions, and generates standardized report templates tailored to the requirements of NIH, Gates Foundation, and Wellcome Trust. This pillar transforms KIPRE's analytical capacity and dramatically reduces time-to-publication.

3.2 Solution Architecture

The system is a layered, container-based platform running entirely on KIPRE-owned, on-premises infrastructure. The foundation is the digitization layer — an open-source laboratory information backbone (SENAITE) for core lab entities, with bespoke services covering the workflows it does not. On top of that foundation, and only after it is live, sits a thin agentic layer (the AI agent) that observes the platform through scoped tools and raises alerts under human oversight. Every component is open-source, self-hostable, and chosen for low-bandwidth, offline-tolerant operation. Each pillar maps to defined components and a specific agentic role:

Pillar	Primary Components	Agentic AI Role
P1 • Inventory	SENAITE inventory + stock service; QR/barcode; PostgreSQL; procurement hooks	Monitor stock vs. physical counts; forecast restock; flag expiry/discrepancies
P2 • Animal Requests	Workflow engine; mobile/Progressive Web App (PWA) forms; Keycloak roles; ISERC approval routing	Pre-screen requests for completeness; route to approvers; surface Service Level Agreement (SLA) breaches
P3 • Work Orders	Barcode-linked orders; offline mobile app + CouchDB sync; chain-of-custody log	Flag samples missing processing windows or showing anomalous readings
P4 • Data Repository	PostgreSQL + MinIO FAIR repository; Role-Based Access Control (RBAC); versioning; validation at capture	Detect cross-record inconsistencies; suggest corrections
P5 • Analytics	Python analytics; dashboards; funder report templates	Flag statistical anomalies; recommend quality control; assemble report drafts

3.3 Open-Source Technology Stack

All components are open-source and self-hostable on a Linux Server (e.g. Ubuntu); licenses are noted so KIPRE retains full ownership and audit rights.

Frontend

Component	Technology	Role
Web app (in-lab)	React + TypeScript PWA	Installable, cached browser app for dashboards, forms, admin
Mobile / field app	Flutter (offline-first)	Barcode scanning, work-order check-ins, field capture
Barcode/QR + offline store	ZXing; SQLite/PouchDB	On-device decoding; local persistence with queued sync

Backend & application services

Component	Technology	Role
LIMS backbone	SENAITE (Plone/Python, GPL-2.0)	Samples, analyses, instrument results, reporting; JSON/REST API
Application services	FastAPI (Python, MIT)	Workflows, agent tool endpoints, integration, business logic
Identity & RBAC	Keycloak (Apache-2.0)	SSO and role-based access control
Async tasks / instruments	Celery + Redis; senaite.astm	Alerts, forecasts, scheduled jobs; analyser feeds

SENAITE note. SENAITE runs on the mature Plone/Zope stack and is actively maintained (2.6.0, with regular releases), but is still completing its migration to full Python 3. The deployment should track current SENAITE releases, budget for the Python 3 transition in maintenance planning, and pin tested versions in CI before each upgrade.

Data & storage

Component	Technology	Role
Relational DB + audit	PostgreSQL 16 + pgAudit	Records, workflow state; immutable audit trail for funder/ISO/GLP
Object/file storage	MinIO	Images, raw assay files, documents
Offline sync; cache	CouchDB; Redis	Field-app replication; caching, queues, sessions
Open-data publishing	InvenioRDM	FAIR dataset release to Zenodo / MalariaGEN (Phase 3)

Analytics, DevOps & observability

Component	Technology	Role
Analytics & notebooks	pandas/NumPy/SciPy/statsmodels; JupyterHub	Parasitaemia curves, drug-efficacy, quality control statistics
Dashboards & reporting	Apache Superset; WeasyPrint/senaite.impress	KPI dashboards; NIH/Gates/Wellcome report templates
Deploy / CI-CD	Docker + Compose (k3s at scale); Gitea/GitLab CE	Reproducible deployment; KIPRE source-code ownership

Monitoring / backups	Prometheus + Grafana + Loki; Restic/Borg	Metrics, alerting, log aggregation; encrypted backups
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3.4 Agentic AI Layer & Human Oversight

The agent runtime is OpenClaw (github.com/openclaw), a self-hosted, MIT-licensed, model-agnostic framework whose Gateway handles a ReAct tool-use loop, sessions, memory and scheduling. Because it is model-agnostic, KIPRE can run both either cloud-based AI models (e.g. Claude Opus), or open-weight models locally (e.g. Gemma 4) so no sample or patient data ever leaves the institution. The chosen model should be a multimodal family that natively handles images and charts (microscopy, histology, parasitaemia plots).

Deployment note — cloud pilot. The malaria pilot is delivered on cloud to validate quickly without up-front capital. For the pilot, the agent model is Claude Opus 4.8 accessed through Google Cloud Platform (GCP) Vertex AI, and the SENAITE / FastAPI / Keycloak stack runs on managed GCP services (Cloud SQL, Cloud Storage) in place of owned servers. The pilot runs on de-identified malaria data under ISERC sign-off, with Vertex AI keeping inference inside KIPRE's GCP project and region and no data used for model training. On-premises — a local Gemma 4 on KIPRE-owned hardware (Section 9.1) — remains a production end-state, reached by migration in Phase 3, if KIPRE so chooses.

Human-in-the-loop guardrails. Agents flag, correlate and recommend; they never autonomously perform a regulated action (ethics approval, sample disposal, result sign-off). Approval gates route every consequential step to a named human, every agent suggestion and human decision is written to the immutable audit trail, and a versioned evaluation harness measures false-positive rates so tuning is evidence-based — consistent with KIPRE's ethics framework and ISERC governance.

4. Pilot Study : Malaria Pilot Program

By mutual agreement, the malaria pilot program at KIPRE will serve as the initial pilot for this system. This decision reflects both the scientific significance of malaria research in East Africa and the operational complexity of the malaria workflow, which makes it an ideal stress-test for the proposed AI infrastructure.

4.1 Why the Malaria Program?

- Malaria remains the leading infectious disease burden in Africa; KIPRE's malaria research has direct continental public health relevance
- The malaria workflow involves multiple, complex handoffs: animal passage/inoculation, blood-stage culture, drug challenge, endpoint sampling, and multi-assay analysis making it a comprehensive test of all five digital pillars
- KIPRE's established malaria research expertise provides a robust baseline against which AI-assisted outputs can be validated
- Strict ISERC ethics governance already in place for malaria studies provides the ideal framework to test AI oversight protocols
- Malaria datasets are rich and multi-modal (parasitemia counts, PCR, ELISA, histology, drug-efficacy curves), maximizing the value of the AI analytics module

4.2 Pilot Scope

The malaria pilot will implement all five digital pillars within the malaria program over a **nine-month** period (Months 1–9 of the roadmap). Specifically, the pilot will:

1. Deploy a digital animal request and ISERC-integrated approval workflow for all malaria studies
2. Implement barcode-tracked inventory management for parasite stocks, blood samples, antimalarial compounds, and PCR reagents
3. Launch a mobile-accessible work-order and sample-collection logging system for malaria program technicians
4. Migrate existing malaria data records to the centralized FAIR-compliant repository
5. Deploy the AI analytics dashboard for automated parasitaemia curves, drug-efficacy analysis, and QC flagging

4.3 Expected Pilot Outcomes

By the end of Month 9, the pilot is expected to demonstrate:

KPI	Target
Sample traceability	+40% improvement vs paper baseline
Data entry error rate	Reduction of ≥50% through automated validation
Reagent wastage	≥30% reduction through AI-forecast restocking
Research efficiency	+25% (sample-to-insight cycle time)
Staff AI literacy	>80% of malaria research and ASD staff trained by Month 9
Audit readiness	100% of malaria records audit-trail compliant for funder reporting

5. Implementation Roadmap

The 18-month roadmap is structured in three phases, each with clear milestones and governance checkpoints. The malaria pilot runs through Phases 1 and 2, with institution-wide scale-up beginning in Phase 3. **Detailed technical workstreams are integrated under each phase below; digitization is completed first and the agentic layer is connected last.**

Phase 1 : Foundation (Months 1–3)

- Full process audit and workflow mapping across all five digital pillars in the malaria program
- Technology stack finalization with Upanzi (open-source, locally deployable, low-bandwidth capable)
- Proposal submitted to ISERC for AI-assisted use of malaria data and samples

- KIPRE ICT Committee and DG sign-off on AI governance framework and data policy
- Joint Technical Working Group (JTWG) constituted: KIPRE AI Champion, ICT team, senior malaria PIs, Upanzi architects
- Baseline KPI measurement for all five pillars in malaria program

Technical workstreams:

- Translate malaria workflow maps into data models, state machines and API contracts for all five pillars.
- Provision the cloud environment (GCP project, Vertex AI access to Claude Opus 4.8, Cloud SQL for PostgreSQL, Cloud Storage); deploy the application stack (SENAITE, FastAPI, Keycloak, Redis, CouchDB) on a GCP VM; stand up dev/staging and CI-CD on KIPRE-owned repositories. On-premises hardware is provisioned at the Phase 3 migration.
- Design Keycloak identity/RBAC and the pgAudit + application audit-logging strategy that underpins the 100% audit-readiness KPI.
- AI governance design: Large Language Model (LLM) selection, AI agent topology, scoped tool boundaries and the human-in-the-loop guardrail/escalation policy — the technical input to the ISERC proposal and DG/ICT sign-off.
- Instrument the six pilot KPIs and capture the paper-process baseline.

Phase 2 : Build & Pilot (Months 4–9)

- Sequential module deployment: Inventory → Animal Requests → Work Orders → Data Repository → Analytics
- Upanzi-led AI literacy training for all malaria research and technical staff
- Barcode/QR system installed and go-live across malaria sample chain
- Historical data migration: malaria records from past 2 years loaded to repository
- AI agent layer tuned to malaria-specific workflows; false-positive rate minimized
- Mid-point KPI review at Month 6; course correction as needed
- Pilot closure report at Month 9: KPI measurement, lessons learned, publication draft

Step A — Digitization:

- Month 4 Inventory (P1): SENAITe inventory + stock service; QR at receipt; low-stock/expiry alerts — the Month-4 quick-win demo.
- Month 5 Animal Requests (P2): digital forms replacing paper; approval routing; ISERC-integrated auditable trail.
- Month 6 Work Orders (P3): barcode-linked orders; offline mobile check-ins with CouchDB sync; digital chain of custody.
- Month 7 Data Repository (P4): FAIR schema; RBAC; versioning; automated backups; validation at capture; full audit logging.
- Month 8 Analytics (P5): Python pipeline for parasitaemia/drug-efficacy; Superset dashboards; funder report templates.

- **Parallel: barcode/QR go-live; Extract, Transform, Load (ETL) of two years of records with a dedicated validation and data-cleaning sprint.**

Step B — Agentic augmentation, after the digital foundation is live (Month 8–9):

- **Connect OpenClaw to the now-live tools one pillar at a time; tune to malaria workflows; run the eval harness to drive down false positives within guardrails.**
- **Provision a training sandbox so Upanzi can train staff on the live-like system (supports the >80% literacy KPI).**

Phase 3 : Scale & Sustain (Months 10–18)

- Expand system to two additional KIPRE programs (e.g. human sample biobank, second preclinical program)
- Institution-wide rollout of all five pillars
- Formal publication of KIPRE-Upanzi learnings in peer-reviewed journal (CMU-Africa co-authorship)
- Submission of joint NIH Fogarty / Wellcome Trust grant for Phase 2 funding using malaria pilot data as preliminary evidence
- Development of KIPRE replication blueprint for peer East African institutions.

Technical workstreams:

- **Multi-program configuration so new programs onboard by configuration, not rebuild; scale hardening (optional k3s, PostgreSQL HA, storage expansion).**
- **Hand-over: full source-code transfer to KIPRE, documentation, runbooks and internal-team training — mitigating post-pilot dependency.**
- **Migrate to on-premises for production (optional): stand up the Section 9.1 hardware, swap Claude Opus 4.8 → local Gemma 4 (a configuration change in OpenClaw), and repatriate Cloud SQL → PostgreSQL and Cloud Storage → MinIO — delivering full data sovereignty and offline operation.**
- **Open-data pipeline via InvenioRDM (Opportunity D); containerized replication blueprint with infrastructure-as-code (Opportunity F).**

6. Roles and Responsibilities

KIPRE Contributes	Upanzi Network Contributes
Real-world malaria workflows and operational context	Agentic AI system design, architecture, and development
Preclinical and human sample datasets	Deployment on open-source, locally-hosted technology stack
Validation environment and physical infrastructure	Technical training and capacity building for KIPRE staff
Data governance (ISERC) and ethics oversight	Security architecture and AI safety guardrails

KIPRE Contributes	Upanzi Network Contributes
AI Champion coordination (Dr. Faith I. Onditi)	Academic publication support and CMU-Africa network
Regulatory and funding compliance expertise	Ongoing system maintenance and support (Year 1)

Governance will be exercised through: (1) the KIPRE AI Champion as central coordinator; (2) the Joint Technical Working Group meeting monthly; (3) the KIPRE ICT and Digitalization Committee for resource and policy decisions; and (4) quarterly briefings to the Director General.

Technical governance, security & data sovereignty. The AI agent runs only on the internal network with least-privilege credentials, reaching data exclusively through scoped, permissioned tools rather than direct database or shell access. Platform controls include Keycloak RBAC/SSO, TLS, secrets management, network segmentation, and an immutable pgAudit + application audit trail. All sample and patient data remain on KIPRE-owned, on-premises hardware. During the cloud pilot, inference uses Claude Opus 4.8 via Vertex AI on de-identified data under ISERC approval, kept within KIPRE's GCP region; optional migration to on-premises (Phase 3) removes the off-premises data flow entirely.

Agent sandboxing & prompt-injection defense. Because the agent runtime is a self-hosted assistant framework repurposed as an institutional agent, it is sandboxed rather than trusted by default: the agent process runs in an isolated container with no host shell, an explicit tool allow-list, and egress restricted to the internal network, and tool calls touching regulated entities (ethics approvals, sample disposal, result sign-off) are blocked at the tool layer and routed to human approval. Multimodal inputs (microscopy images, scanned forms) and retrieved records are treated as untrusted prompt-injection vectors: tool outputs are schema-validated, a hardened runtime (e.g., NVIDIA OpenShell/NemoClaw) is the recommended option for stronger isolation, and the full-size (not aggressively quantized) model is run to reduce injection susceptibility.

7. Additional Strategic Opportunities

Beyond the core five-pillar system, the following opportunities would significantly enrich the partnership and maximize KIPRE's return on this investment:

A. Mobile-First Field Data Capture

Development of a low-bandwidth, offline-capable mobile application for field teams collecting samples in remote or resource-limited settings. Data syncs automatically when connectivity is restored, ensuring continuity of chain-of-custody even outside the main campus. This is particularly relevant for KIPRE's field-based biodiversity and zoonotic disease studies.

Now in scope: delivered by the Flutter offline-first app with CouchDB sync in the core stack (Section 3.3).

B. LIMS Integration (Open-Source)

Integration of the AI layer with an open-source Laboratory Information Management System (LIMS) such as Bika/Senaite would provide full instrument-to-report traceability. This positions KIPRE for ISO 17025 accreditation, a significant differentiator when competing for international research contracts.

Now in scope: SENAITE is adopted as the core LIMS backbone (Section 3.3), with senaite.astm instrument feeds.

C. AI Literacy Curriculum for KIPRE Researchers

A structured, Upanzi-designed three-five day AI literacy bootcamp for KIPRE's research and technical staff. Building internal capability reduces external dependency over time and enables KIPRE to become a training hub for peer institutions across the region.

D. Open-Data Contribution to Global Malaria Research

Anonymized, FAIR-compliant KIPRE malaria datasets published to international repositories such as MalariaGEN would elevate KIPRE's global scientific profile, increase citation impact, and demonstrate commitment to open science, a growing requirement for top-tier funders.

Now in scope: delivered via an InvenioRDM open-data pipeline in Phase 3 (Sections 3.3 and 5).

E. Joint Grant Application

The malaria pilot generates compelling preliminary data for a co-PI grant application between KIPRE and CMU-Africa. Target mechanisms include NIH Fogarty International Centre D43 (research training), Wellcome Trust African Institutions Initiative, and the Gates Foundation Grand Challenges.

F. East Africa Replication Model

Upon successful pilot, the KIPRE system should be packaged as a deployable, documented blueprint for replication at local and continental research institutions. This positions KIPRE as a regional hub and technical lead for AI-enabled biomedical research infrastructure, consistent with KIPRE's vision and Kenya's AI strategy.

Now in scope: the containerized reference deployment with infrastructure-as-code is a Phase-3 deliverable (Section 5).

8. Risk Management

Risk	Mitigation
Slow staff uptake / resistance to change	Upanzi-led training; AI Champion peer mentorship; phased rollout; quick-win demonstration in Month 4
ISERC / regulatory approval delays	Early engagement (Month 1); parallel preparation of ethics amendment; legal review by Month 2
Technology connectivity / infrastructure gaps	Offline-capable modules; locally hosted (not cloud-dependent) architecture; ICT infrastructure audit in Phase 1
Data migration quality from paper records	Structured data-entry validation; dedicated data-cleaning sprint in Phase 2; 2-year historical baseline only
Dependency on Upanzi availability post-pilot	Full source-code ownership by KIPRE; comprehensive documentation; internal KIPRE tech team trained in Year 1
Funding gap for Phase 3 scale-up	Joint grant application submitted by Month 9 using malaria pilot as preliminary data; DG briefed on funding strategy

Reinforcing technical mitigations: scoped agent tool permissions and a false-positive eval harness contain AI risk; the immutable audit trail provides funder/ISO/GLP audit readiness; PWA/CouchDB offline sync and on-

premises hosting address connectivity risk; and source-code ownership plus documentation and runbooks remove post-pilot dependency.

9. Indicative Budget Framework

A detailed budget will be developed jointly with Upanzi following the Phase 1 process audit. The framework below outlines the major cost categories for planning purposes: Indicative hardware specifications and costs are now provided in Sections 9.1–9.2; the malaria pilot is hosted on cloud (see Section 3 and the cloud-pilot budget below), with the on-premises hardware as the production target.

Cloud pilot budget (selected for the malaria pilot). The pilot runs on cloud as a monthly operating cost rather than capital: about \$550/month of GCP services (application VM, Cloud SQL, Cloud Storage, networking) plus Claude Opus 4.8 via Vertex AI at roughly \$430/month in the expected case — about \$980/month, or ~\$8,800 over the nine-month pilot (~\$14,000 including a minimal field-device kit). This validates the full pilot for well under \$15,000 and defers the ~\$60,000 on-premises capital (Sections 9.1–9.2) until after the pilot proves out. Opus 4.8 is billed at \$5 / \$25 per million input / output tokens; usage scales with agent context, so the expected figure should be confirmed against a short metered trial.

Cost Category	Notes
System Development (Upanzi)	AI architecture, module development, deployment — scope and cost to be confirmed following workflow audit
Infrastructure & Hardware	Computers, Printers, Servers, barcode scanners, mobile devices for field capture — may be co-funded with ICT Committee budget (itemized specifications and costing in Sections 9.1–9.2)
Staff Training	Upanzi-led AI literacy bootcamp + ongoing malaria team training (3-5 day workshop + monthly refreshers)
Data Migration	Historical malaria data cleaning and loading sprint — KIPRE staff time + Upanzi data engineering support
ISERC & Regulatory	Ethics amendment filing fees, legal review, documentation
Monitoring & Evaluation	KPI measurement, mid-point review, pilot close-out report, publication costs
Phase 3 Scale-Up (estimated)	To be funded via joint NIH/Wellcome grant; Phase 1–2 costs to be met through institutional and partnership budget

Funding Target	A joint KIPRE × CMU-Africa application to NIH Fogarty International (D43 mechanism) or Wellcome Trust African Institutions Initiative will be prepared using the malaria pilot as preliminary data. Target submission: Month 9.
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9.1 Hardware Specifications (KIPRE-Provided)

This section assumes KIPRE opts to move all cloud infrastructure components to on-premise, after the pilot has proven itself. All servers run Ubuntu Server 26.04 LTS on KIPRE premises. The only specialized component is a single GPU server hosting the local Gemma 4 model: Gemma 4 31B needs ~20 GB of VRAM at 4-bit and ~34 GB at 8-bit, so a single 48 GB professional GPU runs it at 8-bit with headroom; the current-generation equivalent is the RTX PRO 6000 Blackwell (96 GB GDDR7), which doubles VRAM headroom where budget allows. A 24–32 GB GPU at 4-bit is a constrained, budget-limited fallback rather than the default —

heavier quantization cuts context headroom and raises prompt-injection risk for a privileged agent — so the 48 GB card at 8-bit is the recommended configuration.

Component	Specification
AI / inference server	1× RTX 6000 Ada / L40S-class 48 GB GPU (current-gen: RTX PRO 6000 Blackwell, 96 GB); 16-core EPYC/Xeon; 128 GB ECC; 2 TB + 4 TB NVMe — vLLM serving + OpenClaw
Application & database server	16-core CPU; 64 GB ECC; 1 TB NVMe + 4 TB SSD RAID-10 — SENAITE, FastAPI, Keycloak, PostgreSQL, MinIO, Redis, CouchDB
Storage, backup & power	8–12 TB RAID-6 NAS (encrypted nightly backups); 2× online UPS (≥3 kVA); VLAN-segmented gigabit LAN
Field & edge devices	Rugged Android tablets; technician smartphones; 2D barcode scanners; thermal label printers; analyst workstations; Wi-Fi APs

9.2 Indicative Hardware Cost Estimate

Method. Unit prices are indicative figures from Newegg.com (priced June 2026, USD); because the hardware ships to Kenya, each is multiplied by 1.5 as a planning allowance for shipping, insurance and import duty/VAT (landed ≈ 1.5× Newegg). These are planning estimates, not quotations.

Memory & storage volatility (priced June 2026). An AI-driven DRAM and NAND shortage through 2026 has pushed server ECC memory up roughly 2–5× over 2024–25 levels and raised SSD prices materially; high-end 128 GB ECC RDIMMs have been quoted near \$4,000–\$4,700. The memory and storage lines in the totals below have been re-priced to June 2026 levels and remain volatile — they should be re-quoted at the point of procurement, and finished devices that contain memory (tablets, workstations) may also rise.

Category	Newegg (USD)	Landed × 1.5 (USD)
AI / Inference Server	\$16,089	\$24,134
Application & Database Server	\$5,750	\$8,625
Storage, Backup & Power	\$4,280	\$6,420
Network	\$950	\$1,425
Field & Edge Devices	\$13,100	\$19,650
Total estimated hardware	\$40,169	\$60,254

The GPU is the largest single line (RTX 6000 Ada 48 GB ≈ \$7,669 on Newegg), and the increase over earlier estimates is driven mainly by the 2026 server-memory price surge. Consolidating the two servers onto one machine, refurbished workstations, and phasing field-device purchases with the rollout are all viable savings without changing the architecture; a 24–32 GB GPU at 4-bit is the single largest possible saving but — per Section 9.1 — is a constrained, budget-limited interim choice rather than the default. The hardware is a one-time, KIPRE-owned capital investment (with ~\$3,500/year for power, cooling and maintenance) expected to serve 3–5 years, consistent with the locally-hosted, non-cloud-dependent architecture this concept note specifies.

For the malaria pilot, however, the platform is hosted on cloud to validate first. As such, none of the above hardware expenses will be incurred, until KIPRE decides to migrate a fully self-hosted, on-premise infrastructure.

10. Conclusion and Recommended Next Steps

The KIPRE × Upanzi Network partnership represents a transformative opportunity for KIPRE to leapfrog incremental digitalization and establish itself as East Africa’s first AI-enabled biomedical research institute. The five-pillar digital infrastructure, anchored by the malaria pilot, will deliver measurable improvements in research quality, operational efficiency, funding compliance, and institutional reputation.

Recommendation

The Director General is requested to formally endorse the KIPRE × Upanzi collaboration, authorize the AI Champion to constitute the Joint Technical Working Group, and direct the ISERC secretariat to prioritize the ethical approval for the malaria pilot.

Immediate Next Steps (Weeks 1–6)

1. AI Champion shares full malaria process maps with Upanzi team — within 2 weeks
2. ICT Committee briefed on partnership and requested to begin infrastructure audit — within 2 weeks
3. ISERC ethics amendment for AI pilot submitted — within 4 weeks
4. Upanzi delivers technology stack proposal and data architecture draft — within 3 weeks
5. Joint Technical Working Group first meeting convened; Terms of Reference agreed — within 4 weeks
6. MOU / Letter of Intent between KIPRE and CMU-Africa signed — within 6 weeks
7. AI Literacy Orientation for KIPRE staff scheduled — within 2 months
8. Joint NIH Fogarty grant preparation commenced — within 3 months

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